

HIGH WYCOMBE, United Kingdom

WMO: 036600

Lat: 51.6817N

Lon: 0.8069W

Elev: 205

StdP: 98.89

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.9	-6.1	2.3	-0.2	-4.8	2.6	-0.2	9.4	8.0	8.6	8.1	3.1	300	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	26.1	18.2	24.0	17.3	22.3	16.5	19.1	24.3	18.1	22.5	17.2	20.9	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.6	20.6	16.4	12.0	19.6	15.7	11.4	18.7	55.0	24.2	51.8	22.6	49.1	20.9	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	7.1	6.3	-4.5	29.4	1.5	2.6	-5.6	31.2	-6.5	32.8	-7.3	34.3	-8.4	36.2		
			-4.9	21.2	1.6	1.6	-6.1	22.4	-7.0	23.4	-7.9	24.3	-9.0	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	4.3	4.5	6.3	8.7	11.8	14.6	16.7	16.3	14.2	10.8	7.0	4.9
	DBStd	5.34	3.09	3.20	3.13	3.14	2.95	2.66	2.74	2.44	2.46	2.66	2.93	3.39
	HDD10.0	802	178	154	118	59	14	1	0	0	1	21	97	159
	HDD18.3	3058	436	386	372	289	204	116	67	72	126	234	341	416
	CDD10.0	821	1	1	4	21	68	139	209	197	128	46	6	2
	CDD18.3	35	0	0	0	0	1	4	17	11	2	0	0	0
	CDH23.3	253	0	0	0	1	5	39	128	67	11	1	0	0
	CDH26.7	42	0	0	0	0	0	6	24	13	0	0	0	0
Wind	WSAvg	3.2	3.7	3.6	3.6	3.3	3.2	2.9	2.8	2.7	2.9	3.3	3.2	3.7
Precipitation	PrecAvg	672	65	47	41	52	51	50	49	60	55	67	73	62
	PrecMax	878	156	125	99	136	135	137	111	141	115	145	146	112
	PrecMin	495	6	1	12	1	4	11	14	8	12	20	30	13
	PrecStd	117	33	30	22	34	28	30	25	34	31	32	33	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	13.5	16.5	21.7	23.9	27.3	29.2	28.2	25.0	19.4	14.7	12.9
		MCWB	10.9	10.1	10.6	13.7	16.7	18.7	19.4	19.8	17.6	15.7	13.5	11.6
	2%	DB	11.1	11.2	13.9	18.1	21.4	24.0	26.6	25.0	22.1	16.8	13.3	11.8
		MCWB	9.9	9.2	9.7	12.4	15.5	17.2	18.4	18.3	16.3	14.4	12.0	10.9
	5%	DB	10.0	10.0	12.2	15.7	19.3	21.8	24.4	22.7	20.0	15.6	12.3	10.9
		MCWB	8.9	8.5	9.2	11.0	14.3	16.3	17.6	17.0	15.5	13.7	11.2	10.0
	10%	DB	8.9	9.0	10.8	13.7	17.0	19.8	22.3	20.9	18.3	14.6	11.3	9.9
		MCWB	8.0	7.7	8.5	9.9	12.9	15.2	16.7	16.0	14.7	13.1	10.2	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.0	11.7	14.9	18.0	19.8	20.6	20.7	18.9	16.5	13.9	12.1
		MCDB	11.7	12.5	14.2	20.3	21.9	25.9	27.1	27.1	22.4	18.6	14.4	12.4
	2%	WB	10.1	9.7	10.6	12.9	16.1	18.0	19.1	18.8	17.4	15.1	12.4	11.2
		MCDB	10.8	10.7	12.7	17.4	20.1	22.3	24.9	23.3	20.6	16.2	13.0	11.7
	5%	WB	9.1	8.8	9.8	11.6	14.7	16.7	18.1	17.7	16.4	14.2	11.4	10.2
		MCDB	9.7	9.7	11.5	14.7	18.3	20.5	23.1	21.3	18.8	15.1	12.1	10.8
	10%	WB	8.1	7.8	8.9	10.5	13.5	15.7	17.2	16.7	15.5	13.4	10.4	9.0
		MCDB	8.8	8.9	10.6	13.1	16.4	19.3	21.3	19.7	17.6	14.4	11.1	9.8
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.9	6.2	7.6	8.0	8.3	8.6	8.1	7.4	5.7	4.7	4.2
		MCDBR	5.5	5.8	7.9	10.3	10.8	11.4	12.0	10.8	9.4	6.3	5.2	4.8
		MCWBR	5.4	4.4	5.0	5.9	6.2	6.1	5.5	5.2	4.8	4.3	4.6	4.8
	5% WB	MCDBR	5.3	5.6	7.0	9.3	9.8	10.2	10.6	9.3	8.3	5.8	5.1	4.8
		MCWBR	5.4	4.8	4.9	5.7	5.9	5.9	5.5	5.0	5.1	4.6	4.9	5.0
Clear-Sky Solar Irradiance	taub	0.322	0.338	0.372	0.394	0.398	0.397	0.393	0.383	0.366	0.350	0.324	0.310	
	taud	2.373	2.386	2.282	2.237	2.262	2.289	2.307	2.354	2.402	2.430	2.418	2.356	
	Ebn at Noon	683	781	815	841	855	858	855	844	817	755	676	628	
	Edh at Noon	62	80	106	125	127	125	121	110	94	76	59	54	
All-Sky Solar Radiation	RadAvg	0.81	1.47	2.54	3.91	4.76	5.23	4.97	4.12	3.11	1.81	0.96	0.64	
	RadStd	0.07	0.19	0.34	0.45	0.30	0.53	0.50	0.33	0.27	0.16	0.11	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page